Double Integrals

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Double integrals over a rectangle

Set up the double integral of $f(x, y) = xy$ over the rectangle $[0, 2] \times [0, 3]$ (integrate y first).

$\int_0^2 \int_0^3 xy \, dy \, dx$.

2 [Foundational] • Target: Fubini's Theorem: order of integration

Verify Fubini's Theorem for $f(x, y) = xy$ on $[0, 1] \times [0, 2]$ by evaluating the integral in both orders.

Order $dy\,dx$: inner $= x \cdot \frac{y^2}{2}\big|_0^2 = 2x$, outer $= \int_0^1 2x \, dx = 1$. Order $dx\,dy$: inner $= y \cdot \frac{x^2}{2}\big|_0^1 = \frac{y}{2}$, outer $= \int_0^2 \frac{y}{2} \, dy = 1$. Both orders give 1.

3 [Foundational] • Target: Fubini's Theorem: order of integration

State the hypothesis of Fubini's Theorem needed for the two iterated orders to be guaranteed equal.

$f(x, y)$ must be continuous (or at least integrable) on the rectangular region $[a, b] \times [c, d]$ — under that condition, the double integral equals either iterated integral, in either order.

4 [Foundational] • Target: Double integrals over general regions

Set up (do not evaluate) the double integral of $f(x, y) = xy$ over the triangular region bounded by $y = 0$, $x = 1$, and $y = x$.

$\int_0^1 \int_0^x xy \, dy \, dx$ (for a fixed x between 0 and 1, y ranges from the bottom edge $y = 0$ up to the line $y = x$).

5 [Foundational] • Target: Double integrals over general regions

Evaluate $\int_0^1 \int_y^1 2 \, dx \, dy$ (a region bounded by $x = y$, $x = 1$, $y = 0$).

Inner: $\int_y^1 2 \, dx = 2(1 - y)$. Outer: $\int_0^1 2(1 - y) \, dy = 2\left[y - \frac{y^2}{2}\right]_0^1 = 2\left(1 - \frac{1}{2}\right) = 1$.

6 [Foundational] • Target: Reversing the order of integration

Reverse the order of integration for $\int_0^1 \int_0^x f(x, y) \, dy \, dx$, describing the new bounds.

The region is $0 \le y \le x \le 1$. Reversed: for a fixed y between 0 and 1, x ranges from y up to 1: $\int_0^1 \int_y^1 f(x, y) \, dx \, dy$.

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7 [Foundational] • Target: Reversing the order of integration

Evaluate $\int_0^1 \int_x^1 e^{y^2}\,dy\,dx$ by first reversing the order of integration (the inner integral has no elementary antiderivative in the original order).

The region is $0 \le x \le y \le 1$. Reversed: $\int_0^1 \int_0^y e^{y^2}\,dx\,dy = \int_0^1 y\,e^{y^2}\,dy$. Substituting $u = y^2$, $du = 2y\,dy$: $= \frac{1}{2}\int_0^1 e^u\,du = \frac{1}{2}(e-1)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Polar coordinates

Convert $\iint_D (x^2+y^2)\,dA$ to polar coordinates, where D is the disk $x^2+y^2 \le 4$, and evaluate.

$x^2+y^2 = r^2$, $dA = r\,dr\,d\theta$. $\int_0^{2\pi}\int_0^2 r^3\,dr\,d\theta$. Inner: $\frac{r^4}{4}\big|_0^2 = 4$. Outer: $\int_0^{2\pi} 4\,d\theta = 8\pi$.

9 [Intermediate] • Target: Polar coordinates

Evaluate $\iint_D xy\,dA$ where D is the quarter disk $x^2+y^2 \le 1$, $x \ge 0$, $y \ge 0$, using polar coordinates.

$xy = r^2\sin\theta\cos\theta$, $dA = r\,dr\,d\theta$, $\theta \in [0, \frac{\pi}{2}]$, $r \in [0,1]$. $\int_0^{\pi/2} \sin\theta\cos\theta\,d\theta \cdot \int_0^1 r^3\,dr = \frac{1}{2}\cdot\frac{1}{4} = \frac{1}{8}$.

10 [Intermediate] • Target: Polar coordinates

Use polar coordinates to evaluate $\iint_D e^{-(x^2+y^2)}\,dA$ where D is the entire plane $\mathbb{R}^2$ (the classic Gaussian integral setup).

$\int_0^{2\pi}\int_0^\infty e^{-r^2} r\,dr\,d\theta$. Inner: let $u = r^2$, $du = 2r\,dr$: $\int_0^\infty e^{-r^2} r\,dr = \frac{1}{2}\int_0^\infty e^{-u}\,du = \frac{1}{2}$. Outer: $\int_0^{2\pi} \frac{1}{2}\,d\theta = \pi$.

11 [Intermediate] • Target: Area and average value

Use a double integral to find the area of the region bounded by $y = x^2$ and $y = x$ (for $0 \le x \le 1$, where $x \ge x^2$).

Area $= \int_0^1 \int_{x^2}^x 1\,dy\,dx = \int_0^1 (x - x^2)\,dx = \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$.

12 [Intermediate] • Target: Area and average value

The average value of $f(x,y)$ over a region D with area A is $\frac{1}{A}\iint_D f\,dA$. Find the average value of $f(x,y) = xy$ over the rectangle $[0,2] \times [0,3]$, using the earlier result $\iint_D xy\,dA = 9$.

Area of $[0,2]\times[0,3]$ is 6. Average value $= \frac{9}{6} = \frac{3}{2}$.

13 [Intermediate] • Target: Area and average value

Use a double integral to find the area of the region bounded above by $y = 4 - x^2$ and below by $y = 0$.

The parabola meets $y = 0$ at $x = \pm 2$. Area $= \int_{-2}^2 \int_0^{4-x^2} 1\,dy\,dx = \int_{-2}^2 (4 - x^2)\,dx = \left[4x - \frac{x^3}{3}\right]_{-2}^2 = \left(8 - \frac{8}{3}\right) - \left(-8 + \frac{8}{3}\right) = 16 - \frac{16}{3} = \frac{32}{3}$.

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14 **[Intermediate]** • **Target: Volume via double integrals**

Set up (do not evaluate) the double integral for the volume under the surface $z = x^2 + y^2$ above the rectangle $[0, 1] \times [0, 1]$.

$V = \int_0^1 \int_0^1 (x^2 + y^2)\, dy\, dx$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Volume via double integrals** • **Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers**

Evaluate the volume integral set up above.

Inner: $\int_0^1 (x^2 + y^2)\, dy = x^2 y + \frac{y^3}{3}\big|_0^1 = x^2 + \frac{1}{3}$. Outer: $\int_0^1 \left(x^2 + \frac{1}{3}\right) dx = \frac{x^3}{3} + \frac{x}{3}\big|_0^1 = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$.

16 **[Challenging]** • **Target: Volume via double integrals** • **Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives**

Find the volume of the solid bounded above by $z = 6 - x - y$ and below by $z = 0$, over the triangle in the xy-plane with vertices $(0, 0)$, $(6, 0)$, $(0, 6)$ (i.e., $0 \le x \le 6$, $0 \le y \le 6 - x$).

$V = \int_0^6 \int_0^{6-x} (6 - x - y)\, dy\, dx$. Inner: with $u = 6 - x$, $\int_0^u (u - y)\, dy = uy - \frac{y^2}{2}\big|_0^u = u^2 - \frac{u^2}{2} = \frac{u^2}{2}$. Outer: $\int_0^6 \frac{(6-x)^2}{2}\, dx$. Substituting $u = 6 - x$, $du = -dx$: $= \int_0^6 \frac{u^2}{2}\, du = \frac{u^3}{6}\big|_0^6 = \frac{216}{6} = 36$.

17 **[Challenging]** • **Target: Double integrals in applications: mass and center of mass** • **Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives**

A flat plate occupies the rectangle $[0, 2] \times [0, 1]$ with density $\rho(x, y) = x + y$. Set up the double integral for its total mass $M = \iint_D \rho\, dA$.

$M = \int_0^2 \int_0^1 (x + y)\, dy\, dx$.

18 **[Challenging]** • **Target: Double integrals in applications: mass and center of mass** • **Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers**

Evaluate the mass integral set up above.

Inner: $\int_0^1 (x + y)\, dy = xy + \frac{y^2}{2}\big|_0^1 = x + \frac{1}{2}$. Outer: $\int_0^2 \left(x + \frac{1}{2}\right) dx = \frac{x^2}{2} + \frac{x}{2}\big|_0^2 = 2 + 1 = 3$. So $M = 3$.

19 **[Challenging]** • **Target: Double integrals in applications: mass and center of mass** • **Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives**

The x-coordinate of the center of mass is $\bar{x} = \frac{1}{M} \iint_D x\rho\, dA$. Set up (do not evaluate) this integral for the plate above ($\rho(x, y) = x + y$ on $[0, 2] \times [0, 1]$, $M = 3$).

$\bar{x} = \frac{1}{3} \int_0^2 \int_0^1 x(x + y)\, dy\, dx$.

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20 [Challenging] • Target: Double integrals in applications: mass and center of mass • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Evaluate $\bar{x}$ from the integral set up above.

Inner: $\int_0^1 x(x+y)\,dy = \int_0^1 (x^2+xy)\,dy = x^2y + \frac{xy^2}{2}\big|_0^1 = x^2 + \frac{x}{2}$. Outer: $\int_0^2 \left(x^2+\frac{x}{2}\right)dx = \frac{x^3}{3} + \frac{x^2}{4}\big|_0^2 = \frac{8}{3} + 1 = \frac{11}{3}$. So $\bar{x} = \frac{1}{3}\cdot\frac{11}{3} = \frac{11}{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Double integrals over a rectangle

The figure shows the region R bounded by the curves $y = x^2$ and $y = 4$. Which of the following double integrals represents the area of region R using the order of integration $dydx$?

a. $\int_{-2}^{2}\int_{x^2}^{4} dydx$ ✓

b. $\int_{-2}^{2}\int_{4}^{x^2} dydx$

c. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dydx$

d. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dxdy$

The region R is bounded above by $y = 4$ and below by $y = x^2$. To find the limits for x, we find the intersection points of the two curves: $x^2 = 4 \Rightarrow x = \pm 2$. Thus, the region is defined for x from -2 to 2. When integrating with respect to y first ($dydx$), the inner integral goes from the lower curve to the upper curve, so from $y = x^2$ to $y = 4$. The outer integral is over the range of x, which is -2 to 2. Therefore, the double integral representing the area is.

22 [Foundational] • Target: Iterated integrals

Use cylindrical coordinates to find the volume of the cone $\frac{2}{4}\sqrt{x^2+y^2} \le z \le 2$.

a. $\frac{32\pi}{3}$ ✓

b. 32π

c. 16π

d. $\frac{8\pi}{3}$

$V = \int_0^{2\pi}\int_0^4\int_{\frac{1}{2}r}^{2} r\,dz\,dr\,d\theta = \frac{32\pi}{3}$.

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23 [Foundational] • Target: Fubini's Theorem: order of integration

Volume under $z = 1$ over region R of area A is:

a. A^2

b. A ✓

c. $A/2$

d. πA

Applying Applications of Integration principles, $V = \iint_R 1 \, dA = A$. Therefore, the verified correct option is A.

24 [Foundational] • Target: Double integrals over general regions

Fubini's theorem allows switching the order of integration for:

a. all functions

b. only constant functions

c. **non-negative or absolutely integrable functions** ✓

d. periodic functions

Fubini: valid for non-negative or $f \in L^1$.

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25 [Foundational] • Target: Reversing the order of integration

Consider the region bounded by the curves $y = x^2$ and $y = 4$. This region is revolved around the line $y = 5$ to form a solid. A cylindrical hole is drilled through the center of this solid, perpendicular to the axis of revolution. The radius of the cylindrical hole is r. The volume of the material removed by the cylindrical hole is given by $V_{\text{hole}} = \pi\int_a^b (R(x)^2 - r^2)dx$, where $R(x)$ is the outer radius of the solid at a given x and a, b are the limits of integration. Find the value of a and b, and the expression for V_{hole} in terms of r. The question asks for the setup of the integral representing the volume of the material removed.

a. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((5 - x^2)^2 - r^2)dx$ ✓

b. Limits of integration are 0 and 4; $V_{\text{hole}} = \pi\int_0^4((4 - x^2)^2 - r^2)dx$

c. Limits of integration are 1 and 4; $V_{\text{hole}} = \pi\int_1^4((5 - x^2)^2 - r^2)dx$

d. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((4 - x^2)^2 - r^2)dx$

The region is bounded by $y = x^2$ and $y = 4$. To find the intersection points, we set $x^2 = 4$, which gives $x = \pm 2$. These are our limits of integration, $a = -2$ and $b = 2$. The region is revolved around the line $y = 5$. The outer radius $R(x)$ is the distance from the axis of revolution ($y = 5$) to the outer boundary of the region. The outer boundary is the curve $y = x^2$. So, $R(x) = 5 - x^2$. The inner radius of the cylindrical hole is given as r. The volume of the material removed by the cylindrical.

26 [Foundational] • Target: Polar coordinates

Find the volume of the solid between the paraboloid $z = x^2 + y^2$ and the plane $z = 1$.

a. π

b. $\frac{\pi}{4}$

c. $\frac{3\pi}{2}$

d. $\frac{\pi}{2}$ ✓

In cylindrical coordinates, $V = \int_0^{2\pi}\int_0^{\sqrt{1}}(1 - r^2)\,r\,dr\,d\theta = \frac{\pi}{2}$.

27 [Foundational] • Target: Area and average value

Use $A = \oint_C x\,dy$ to find the area of the triangle with vertices $(0,\ 0)$, $(4,\ 0)$, $(0,\ 5)$.

a. 20

b. $\frac{20}{3}$

c. $\frac{9}{2}$

d. 10 ✓

Only the hypotenuse contributes to $\oint x\,dy$; along it $\int x\,dy = 10$. The area is 10.

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28 [Foundational] • Target: Volume via double integrals

Find the volume of the tetrahedron bounded by the coordinate planes and $x + y + z = 3$.

a. $\frac{9}{2}$ ✓

b. 9

c. $\frac{9}{4}$

d. $\frac{27}{2}$

$\int_0^3 \int_0^{3-x} \int_0^{-x-y+3} 1 \, dz \, dy \, dx = \frac{9}{2}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Double integrals in applications: mass and center of mass

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x, y, z \rangle$ across the closed cylinder $x^2 + y^2 \le 1, \ 0 \le z \le 1$.

a. 6π

b. 3π ✓

c. $\frac{3\pi}{2}$

d. -3π

$abla \cdot \mathbf{F} = 3$. The flux equals $\iiint_E abla \cdot \mathbf{F} \, dV = 3\pi$.

30 [Intermediate] • Target: Double integrals over a rectangle

Use $A = \frac{1}{2} \oint_C x \, dy - y \, dx$ to find the area enclosed by $x = 5\cos t, \ y = 2\sin t, 0 \le t \le 2\pi$.

a. 20π

b. 7π

c. 5π

d. 10π ✓

$x \, dy - y \, dx = 10(\cos^2 t + \sin^2 t) \, dt$, so $A = \frac{1}{2} \cdot 10 \cdot 2\pi = 10\pi$.

31 [Intermediate] • Target: Iterated integrals

Find the volume of the solid under $z = 4x + y$ above the unit square $[0, 1] \times [0, 1]$.

a. $\frac{5}{2}$ ✓

b. 5

c. $\frac{5}{4}$

d. $\frac{15}{2}$

$\int_0^1 \int_0^1 \int_0^{4x+y} 1 \, dz \, dy \, dx = \frac{5}{2}$.

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32 [Intermediate] • Target: Fubini's Theorem: order of integration

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x+y, y+z, x+z\rangle$ across the cube $[0,2]^3$.

a. 48

b. 12

c. 24 ✓

d. -24

$abla \cdot \mathbf{F} = 3$. The flux equals $\iiint_E abla \cdot \mathbf{F}\, dV = 24$.

33 [Intermediate] • Target: Double integrals over general regions

A solid cylinder $x^2+y^2 \le 9,\ 0 \le z \le 4$ has density $\delta = \sqrt{x^2+y^2}$. Find its mass.

a. 36π

b. 72π ✓

c. 144π

d. 108π

$m = \int_0^{2\pi}\int_0^4\int_0^3 r \cdot r\, dr\, dz\, d\theta = 72\pi$.

34 [Intermediate] • Target: Reversing the order of integration

Evaluate $\int_0^2\int_y^2 e^{x^2}\, dx\, dy$ by reversing the order of integration.

a. $-\frac{1}{2} + \frac{e^4}{2}$ ✓

b. $-1 + e^4$

c. $\frac{1}{2} + \frac{e^4}{2}$

d. $-\frac{1}{4} + \frac{e^4}{4}$

The inner integral has no elementary antiderivative as written. Reversing the order gives $\int_0^2\int_0^x e^{x^2}\, dy\, dx$, which evaluates to $-\frac{1}{2} + \frac{e^4}{2}$.

35 [Intermediate] • Target: Polar coordinates

Iterated integral over tetrahedron $x+y+z \le 1, x, y, z \ge 0$: volume = 1/6. The integral $\iiint x\, dV =$

a. 1/6

b. 1/24 ✓

c. 1/12

d. 1/8

By symmetry $\iiint x = \iiint y = \iiint z$; and $\iiint (x+y+z) = 3\iiint x$; by integration $\iiint (x+y+z)dV = 1/8$... actually $\iiint x\, dV = 1/24$ (standard result).

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36 [Intermediate] • Target: Area and average value

Pappus's theorem: volume of solid of revolution = $2\pi\bar{x} \cdot A$ where $\bar{x}$ is the centroid of the region and A is the:

a. arc length

b. surface area

c. **area of the region ✓**

d. cross-sectional area

Applying Applications of Integration principles, Pappus's centroid theorem. Therefore, the verified correct option is area of the region.

37 [Intermediate] • Target: Volume via double integrals

Fubini's theorem allows interchanging iterated integrals when:

a. both integrals converge

b. f is continuous

c. $\iint |f| < \infty$ ✓

d. $f \geq 0$

Applying Fundamental Theorem principles, Absolute integrability. Therefore, the verified correct option is $\iint |f| < \infty$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Double integrals in applications: mass and center of mass • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Find the volume when the area between $y = \sec x$ and $y = \cos x$ (from $-\pi/4$ to $\pi/4$) is rotated about the x-axis:

a. $\pi^2/4$

b. $2\pi(\pi/4 - 1/2)$

c. $\pi(2 - \pi/2)$ ✓

d. $\pi(\sqrt{2} - 1)$

Applying Applications of Integration principles, Washer method. Therefore, the verified correct option is $\pi(2 - \pi/2)$.

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39 [Challenging] • Target: Double integrals over a rectangle • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Moments of inertia: second moment of area $I_x = \int y^2\,dA$ for rectangle $b \times h$ about its centroidal axis:

a. $bh^3/12$ ✓

b. $bh^3/3$

c. $bh^2/12$

d. $b^2h/12$

Applying Applications of Integration principles, $I_x = bh^3/12$. Therefore, the verified correct option is $bh^3/12$.

40 [Challenging] • Target: Iterated integrals • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

For polar curve $r = e^{\theta}$, $\theta \in [0, 2\pi]$, the area swept is:

a. $e^{4\pi}$

b. $\frac{1}{2}(e^{4\pi} - 1)$

c. $\frac{1}{4}(e^{4\pi} - 1)$ ✓

d. $\pi e^{4\pi}$

$A = \frac{1}{2}\int_0^{2\pi} e^{2\theta} d\theta = \frac{1}{4}[e^{2\theta}]_0^{2\pi} = \frac{e^{4\pi}-1}{4}$.

41 [Challenging] • Target: Fubini's Theorem: order of integration • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Area of disk $x^2 + y^2 \le R^2$ by polar coordinates:

a. πR

b. $2\pi R$

c. πR^2 ✓

d. $4\pi R^2$

$\int_0^{2\pi}\int_0^{R} r\,dr\,d\theta = 2\pi \cdot R^2/2 = \pi R^2$.

42 [Challenging] • Target: Double integrals over general regions • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Area of surface $z = \sqrt{4 - x^2 - y^2}$ over disk $x^2 + y^2 \le 4$:

a. 4π

b. π

c. 8π ✓

d. 2π

Upper hemisphere radius 2: surface area $= 2\pi(2)^2 = 8\pi$.

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43 [Challenging] • Target: Reversing the order of integration • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Average value of $f(x) = \sin x$ over $[0, \pi]$ is:

a. 0

b. $2/\pi$ ✓

c. $1/\pi$

d. $\pi/2$

$\frac{1}{\pi}\int_0^{\pi} \sin x\, dx = \frac{2}{\pi}$.

44 [Challenging] • Target: Polar coordinates • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Volume integral: using spherical coordinates, $dV =$

a. $dr\, d\theta\, d\phi$

b. $r^2 \sin\phi\, dr\, d\theta\, d\phi$ ✓

c. $r\sin\theta\, dr\, d\theta\, d\phi$

d. $r^2\, dr\, d\theta\, d\phi$

Spherical: $dV = r^2 \sin\phi\, dr\, d\theta\, d\phi$.

45 [Challenging] • Target: Area and average value • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Area of surface of revolution $r = f(\theta)$ about polar axis:

a. $2\pi\int f\sin\theta\sqrt{f^2 + f'^2}\, d\theta$ ✓

b. $\pi\int f^2\, d\theta$

c. $2\pi\int f\, d\theta$

d. $4\pi\int f\cos\theta\, d\theta$

Surface area in polar coordinates.

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